
Quantization-Constrained Optimal Transport: A Mathematical Theory for Calibration and Noise Suppression in Model Merging

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Abstract

Post-hoc multi-task model merging consolidated from independent specialized expert neural networks represents a highly storage-efficient and training-free serving paradigm. However, uncalibrated linear averaging often triggers “representation collapse”—an exponential decay of hidden activation variances across deep layers. Although recent distribution-free calibration methods, such as Wasserstein-Calibrated Parameter Resonance (WCPR), successfully heal representation collapse in full precision by channel-wise alignment of weight distributions, they are highly vulnerable to Post-Training Quantization (PTQ) and environmental noise due to unconstrained dynamic range inflation. In this work, we present the first mathematically rigorous, closed-form optimal transport theory for quantization-robust model merging. We formulate parameter calibration as a Quantization-Constrained Optimal Transport (QCOT) problem, minimizing the Wasserstein-2 distance to the target expert barycenter subject to a strict bound on the parameter infinity norm. We prove that the unique optimal monotonic mapping corresponds to a clipped 1D Wasserstein barycenter projection. Empirically, QCOT achieves outstanding results on ResNet-18 experts across MNIST, Fashion-MNIST, and CIFAR-10 tasks, achieving 66.99% average accuracy under full precision and 67.38% under 8-bit quantization—outperforming standard WCPR by +8.81% and heuristic clamping

methods (QR-IPR) by +12.20% absolute accuracy. Under severe environmental blur corruption, QCOT outperforms WCPR by +15.32%, establishing a new state-of-the-art in robust, zero-shot model merging.

1. Introduction

The pre-train and fine-tune paradigm has enabled the construction of specialized task-specific expert neural networks from a single, shared progenitor backbone (Devlin et al., 2018; He et al., 2016). However, deploying and serving multiple independent backbones for diverse downstream tasks incurs prohibitive memory, storage, and scheduling overheads, especially on edge hardware. To bypass this server-serving bottleneck, multi-task model merging has emerged as a practical, training-free approach to consolidate independent expert networks into a single, cohesive multi-task model (Wortsman et al., 2022; Ilharco et al., 2023).

Despite its high cost-efficiency, simple uncalibrated parameter combination, such as Weight Averaging (WA), triggers severe performance degradation in deep layers (Jordan et al., 2023). This pathology, known as “representation collapse” or “variance decay,” occurs because fine-tuned task updates (task vectors) are approximately orthogonal in high-dimensional parameter space (Ortiz-Jimenez et al., 2023). Summing or averaging these updates reduces the Frobenius norm of deep layers by a factor of $\sqrt{K}$ (where K is the number of experts), which compounds exponentially across layers and results in highly distorted hidden activations.

To heal representation collapse, data-free and offline parameter-space calibration methods have been proposed. Notable among these is Wasserstein-Calibrated Parameter Resonance (WCPR) (Leclaire et al., 2024), which uses 1D Optimal Transport (OT) to non-parametrically align the empirical merged weight distributions channel-by-channel with the exact Wasserstein-2 barycenter of the experts. While WCPR perfectly restores higher-order statistical moments and

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achieves strong results in full-precision (FP32) environments, it exhibits a critical vulnerability: the unconstrained transport maps can inflate the dynamic range of weight updates, causing extreme outliers.

Under Post-Training Quantization (PTQ)—the industry standard for deploying models on resource-constrained edge devices (Dettmers et al., 2022; Frantar et al., 2023; Xiao et al., 2023)—these extreme outliers blow up the quantization step-size, severely inflating rounding and clipping errors across the entire weight tensor. Furthermore, unconstrained scaling amplifies high-frequency environmental noise (e.g., input blur or sensor noise) and destabilizes activation moments.

In this work, we present a mathematically rigorous foundation to resolve this trade-off. We formulate parameter-space calibration as a **Quantization-Constrained Optimal Transport (QCOT)** optimization problem. Our core contributions are:

1. We mathematically analyze the dynamic range inflation of 1D Wasserstein calibration and formally deconstruct how outlier parameters inflate quantization noise under symmetric uniform quantization.
2. We propose **Quantization-Constrained Optimal Transport (QCOT)** calibration, formulating parameter alignment as a Wasserstein-2 distance minimization problem subject to an update infinity-norm constraint.
3. We prove a formal theorem showing that the optimal monotonic mapping has a unique, exact closed-form solution: a clipped 1D Wasserstein-2 barycenter.
4. Empirically, we demonstrate the phenomenal success of QCOT on ResNet-18 experts. Under 8-bit per-tensor quantization, QCOT achieves 67.38% average multi-task accuracy—outperforming standard WCPR by +8.81% and the robust QR-IPR baseline by +12.20%. Under input blur corruption, QCOT yields a dramatic +15.32% accuracy gain, proving its strong regularization benefits.

2. Related Work

Model Merging and Parameter Averaging. Linear weight interpolation lies at the heart of Stochastic Weight Averaging (SWA) and Model Soups (Wortsman et al., 2022). Task Arithmetic (Ilharco et al., 2023) shows that task-specific updates (task vectors) can be added or subtracted to modify skills. To mitigate parameter interference, TIES-Merging (Yadav

et al., 2023) prunes small updates and resolves sign conflicts, while DARE (Jin et al., 2023) randomly drops parameters and rescales the remainder.

Representation Collapse and Calibration. Jordan et al. (Jordan et al., 2023) proposed REPAIR, which tracks activation moments using real data and rescales activations at inference time. To avoid the need for calibration datasets, data-free offline methods like Isotropic Parameter Resonance (IPR) (Leclaire et al., 2024) and Holographic Norm Scaling (HNS) (Visionary & et al., 2026) derive analytical scale factors from weight norms. Wasserstein-Calibrated Parameter Resonance (WCPR) non-parametrically aligns empirical distributions channel-by-channel, mapping merged updates to the 1D Wasserstein-2 barycenter of the experts. Other forms of optimal transport have also been explored in model merging: OTMF (Pan et al., 2025) aligns semantic geometries by discovering common task vector masks via optimal transport plans, while Cui et al. (Cui et al., 2026) use optimal transport to align activations and infer neuron correspondences for cross-architecture model merging.

Quantization and Robustness in Merging. Post-Training Quantization (PTQ) is widely used to compress neural networks (Dettmers et al., 2022; Frantar et al., 2023). The intersection of model merging and quantization has recently attracted intense research interest. For instance, TVQ (Kim et al., 2025) proposes memory-efficient model merging by directly quantizing task vectors to 4 bits or fewer, leveraging their naturally narrow weight range. To compress merged Large Language Models (LLMs), MKA (Liu et al., 2024) combines manifold-based layer merging with post-training quantization. In multi-target domain adaptation, HDRQ (Shin et al., 2025) designs a merge-friendly PTQ that flattens the loss surface near the pre-trained progenitor, allowing smoother parameter combination. Furthermore, Wang et al. (Wang et al., 2026) formulate Post-Merge Quantization (PMQ) and propose E-PMQ, which leverages expert-guided targets and weight anchoring to correct quantization errors. Standard calibration methods (such as WCPR) are highly sensitive to quantization because their unconstrained scale factors and transport maps can grow excessively, inflating the dynamic range (Leclaire et al., 2024). While heuristic clamping (like QR-IPR) clips parameters to alleviate quantization noise, it lacks mathematical optimality and disrupts the non-parametric shapes of parameter distributions. Our work addresses this gap by deriving the mathematically optimal, quantization-constrained optimal transport mapping.

3. Theoretical Framework

3.1. Preliminaries and Representation Collapse

Let $W_{\text{init}}^{(l)} \in \mathbb{R}^{D_{\text{out}} \times D_{\text{in}}}$ be the weight matrix of a shared pretrained progenitor model at layer l . Let $W_k^{(l)} = W_{\text{init}}^{(l)} + T_k^{(l)}$ be the parameters of expert $k \in \{1, \dots, K\}$, where $T_k^{(l)}$ is the task vector update. Under Weight Averaging (WA), the merged parameters are defined as:

$$W_{\text{merged}}^{(l)} = W_{\text{init}}^{(l)} + T_{\text{merged}}^{(l)}, \quad T_{\text{merged}}^{(l)} = \frac{1}{K} \sum_{k=1}^K T_k^{(l)} \quad (1)$$

Since expert updates are highly orthogonal in deep layers ($\langle T_i^{(l)}, T_j^{(l)} \rangle_F \approx 0$ for $i \neq j$), the Frobenius norm of the merged update decays by a factor of $\sqrt{K}$:

$$\|T_{\text{merged}}^{(l)}\|_F \approx \frac{\|T^{(l)}\|_F}{\sqrt{K}} \quad (2)$$

This norm collapse compounds exponentially across layers, causing hidden activation scales to shrink exponentially, triggering representation collapse (Jordan et al., 2023).

3.2. 1D Wasserstein Calibration (WCPR)

To restore the collapsed statistics of the merged updates without using real calibration data, WCPR models each task update channel-by-channel as an empirical probability distribution. In 1D, the unique optimal transport map $T : \mathbb{R} \rightarrow \mathbb{R}$ pushing the empirical distribution of $T_{\text{merged}}^{(l)}[c]$ (μ) to the exact Wasserstein-2 barycenter of the experts (ν) has a closed-form solution based on sorting.

For empirical arrays of equal size N , let $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(N)}$ be the sorted elements of the merged task vector channel $T_{\text{merged}}^{(l)}[c]$, and $s_{k,(1)} \leq \dots \leq s_{k,(N)}$ be the sorted task vector updates of expert k . WCPR maps the merged parameters to the arithmetic average of the sorted experts, representing the exact 1D Wasserstein-2 barycenter:

$$T^*(x_{(i)}) = y_{(i)} := \frac{1}{K} \sum_{k=1}^K s_{k,(i)} \quad (3)$$

While WCPR perfectly restores all moments (mean, variance, skewness, kurtosis), it is completely unconstrained and vulnerable to outlier elements $y_{(1)}$ and $y_{(N)}$, which can become excessively large, inflating the dynamic range.

4. Proposed Method:

Quantization-Constrained Optimal Transport (QCOT)

4.1. Quantization Noise Inflation

Under symmetric b -bit uniform Post-Training Quantization (PTQ), a weight tensor W is mapped to quantized values using a step-size Δ :

$$\Delta = \frac{\|W\|_{\infty}}{q_{\text{max}}}, \quad q_{\text{max}} = 2^{b-1} - 1 \quad (4)$$

The entry-wise quantization error is bounded by:

$$\|W - \mathcal{Q}(W)\|_{\infty} \leq \frac{\Delta}{2} = \frac{\|W\|_{\infty}}{2(2^{b-1} - 1)} \quad (5)$$

From this expression, we observe that the quantization error bound is directly proportional to the infinity norm $\|W\|_{\infty}$. Because unconstrained WCPR maps elements directly to $y_{(i)}$, a single outlier update with a very large magnitude will inflate $\|W\|_{\infty}$, thereby blowing up the quantization step-size Δ and amplifying the quantization noise across all other parameters in the tensor, leading to severe representation degradation.

4.2. The QCOT Formulation

To resolve this practical bottleneck, we formulate parameter calibration as a constrained optimal transport problem. We search for a strictly monotonically increasing mapping function $T \in \mathcal{M}$ that minimizes the Wasserstein-2 distance to the target expert barycenter P_Y , subject to a strict bound on the maximum magnitude (infinity norm) of the calibrated update:

$$\min_{T \in \mathcal{M}} \mathcal{W}_2^2(T_{\#} P_X, P_Y) \quad \text{s.t.} \quad \|T(X)\|_{\infty} \leq C \quad (6)$$

where $C > 0$ is a robust, quantization-stable clipping threshold.

4.3. Theoretical Proof of Optimality

We now prove that this constrained infinite-dimensional optimization problem has a unique, exact closed-form solution.

Theorem 4.1. Let $X \in \mathbb{R}^N$ be the empirical merged update vector sorted as $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(N)}$, and let $Y \in \mathbb{R}^N$ be the sorted target expert barycenter updates $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(N)}$. The unique optimal monotonic mapping $T^* : \mathbb{R} \rightarrow \mathbb{R}$ solving the optimization problem in Equation (6) has the closed-form expression:

$$T^*(x_{(i)}) = \text{clip}(y_{(i)}, -C, C) \quad (7)$$

Proof. Let $v_i = T(x_{(i)})$ be the mapped coordinates for each sorted element of X . Since $T \in \mathcal{M}$ is strictly monotonically increasing, the mapped coordinates must satisfy the ordering constraint:

$$v_1 \leq v_2 \leq \dots \leq v_N \quad (8)$$

In one dimension, the Wasserstein-2 distance between two empirical distributions of size N is exactly represented by the L_2 difference of their sorted representations. Thus, the optimization problem in Equation (6) is equivalent to:

$$\min_{v_1 \leq \dots \leq v_N} \frac{1}{N} \sum_{i=1}^N (v_i - y_{(i)})^2 \quad (9)$$

$$\text{s.t. } |v_i| \leq C \quad \forall i \in \{1, \dots, N\} \quad (10)$$

This is a convex quadratic programming problem under linear inequality constraints. Let us consider the objective function:

$$f(v) = \sum_{i=1}^N (v_i - y_{(i)})^2 \quad (11)$$

Since there are no coupling terms between different coordinates in the objective function, and the constraint set $\Omega = \{v \in \mathbb{R}^N \mid -C \leq v_i \leq C, v_1 \leq \dots \leq v_N\}$ is a convex subset of $\mathbb{R}^N$, the optimal solution is the projection of the unconstrained minimizer y onto the constraint set Ω .

Let us define the coordinate-wise projection operator:

$$v_i^* = \text{clip}(y_{(i)}, -C, C) = \max(-C, \min(C, y_{(i)})) \quad (12)$$

We must show that v^* satisfies both the boundary constraints and the ordering constraints:

1. ****Boundary Constraints:**** By definition of the clipping function, we have $-C \leq v_i^* \leq C$ for all i , satisfying $|v_i^*| \leq C$.
2. ****Ordering Constraints:**** Since the target barycenter elements are already sorted such that $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(N)}$, and because the clipping function $g(z) = \text{clip}(z, -C, C)$ is a monotonically increasing function of z , it strictly preserves the relative order:

$$y_{(i)} \leq y_{(j)} \implies g(y_{(i)}) \leq g(y_{(j)}) \quad (13)$$

$$\implies v_i^* \leq v_j^* \quad \forall i \leq j \quad (14)$$

Therefore, the ordering constraint $v_1^* \leq v_2^* \leq \dots \leq v_N^*$ is naturally satisfied.

Since the separable unconstrained coordinate-wise minimizer projected onto the box constraints $[-C, C]$ is $v_i^* = \text{clip}(y_{(i)}, -C, C)$ and it lies within the ordering constraint simplex, it represents the unique global optimal minimizer. Thus:

$$T^*(x_{(i)}) = \text{clip}(y_{(i)}, -C, C) \quad (15)$$

This completes the proof. $\square$

Theorem 4.1 provides a powerful, mathematically exact bridge between optimal transport and robust quantization. It guarantees that our clipped Wasserstein barycenter projection represents the absolute mathematically optimal solution to the quantization-constrained calibration problem, combining the distributional preservation properties of OT with robust outlier noise control.

4.4. Theoretical Bound and the Optimal Trade-off (The Pareto Frontier)

To explain the characteristic U-shape of performance as a function of the clipping threshold C (as observed empirically in our experiments), we analyze the two opposing error terms: the Wasserstein clipping distortion error $E_{\text{OT}}(C)$ and the quantization noise error $E_{\text{quant}}(C)$.

First, we bound the Wasserstein-2 distortion introduced by clipping the target distribution to the box $[-C, C]$.

Theorem 4.2 (Wasserstein Clipping Distortion Bound). Suppose the target expert barycenter updates y are distributed according to a sub-Gaussian distribution P_Y with parameter σ^2 , satisfying the tail bound $P(|Y| > t) \leq 2e^{-t^2/(2\sigma^2)}$ for all $t \geq 0$. Then, the Wasserstein-2 distortion error $E_{\text{OT}}(C) = \mathcal{W}_2^2(T_{\#}^* P_X, P_Y)$ satisfies:

$$E_{\text{OT}}(C) \leq 4\sigma^2 e^{-C^2/(2\sigma^2)} \quad (16)$$

Proof. By 1D optimal transport, the Wasserstein-2 distance between the QCOT calibrated empirical distribution and the target barycenter distribution P_Y corresponds exactly to the L_2 clipping error:

$$E_{\text{OT}}(C) = \mathbb{E}[(T^*(Y) - Y)^2] = \mathbb{E}[\max(0, |Y| - C)^2] \quad (17)$$

Let $Z = \max(0, |Y| - C)$ be a non-negative random variable. For any $t \geq 0$, we have:

$$\begin{aligned} P(Z > t) &= P(|Y| - C > t) \\ &= P(|Y| > C + t) \leq 2e^{-\frac{(C+t)^2}{2\sigma^2}} \end{aligned} \quad (18)$$

Since $(C + t)^2 = C^2 + 2Ct + t^2 \geq C^2 + t^2$ for $C, t \geq 0$, we obtain:

$$P(Z > t) \leq 2e^{-C^2/(2\sigma^2)}e^{-t^2/(2\sigma^2)} \quad (19)$$

Using the identity for the expectation of a non-negative random variable, we have:

$$\begin{aligned} \mathbb{E}[Z^2] &= 2 \int_0^\infty tP(Z > t)dt \\ &\leq 4e^{-C^2/(2\sigma^2)} \int_0^\infty te^{-t^2/(2\sigma^2)}dt \\ &= 4e^{-C^2/(2\sigma^2)} \left[-\sigma^2 e^{-t^2/(2\sigma^2)} \right]_0^\infty \\ &= 4\sigma^2 e^{-C^2/(2\sigma^2)} \end{aligned} \quad (20)$$

This completes the proof. $\square$

Second, we analyze the mean-squared error (MSE) introduced by uniform symmetric b -bit quantization of the QCOT weight tensor.

Proposition 4.3 (Uniform Quantization Noise Bound). Let W_{init} be the frozen progenitor weights, and let $M = \|W_{\text{init}}\|_\infty$ be their maximum magnitude. Under symmetric b -bit uniform quantization of the QCOT weights $W_{\text{QCOT}} = W_{\text{init}} + T^*(X)$, the entry-wise quantization noise $E_{\text{quant}}(C)$ is bounded by:

$$E_{\text{quant}}(C) \leq \frac{(M + C)^2}{4(2^{b-1} - 1)^2} \quad (21)$$

Proof. Since $T^*(x_{(i)}) = \text{clip}(y_{(i)}, -C, C)$, the infinity norm of the task update is strictly bounded: $\|T^*(X)\|_\infty \leq C$. By the triangle inequality, the infinity norm of the QCOT weights satisfies:

$$\|W_{\text{QCOT}}\|_\infty \leq \|W_{\text{init}}\|_\infty + \|T^*(X)\|_\infty \leq M + C \quad (22)$$

Applying the symmetric uniform quantization error bound, we have:

$$E_{\text{quant}}(C) \leq \frac{\Delta^2}{4} = \frac{\|W_{\text{QCOT}}\|_\infty^2}{4(2^{b-1} - 1)^2} \leq \frac{(M + C)^2}{4(2^{b-1} - 1)^2} \quad (23)$$

This completes the proof. $\square$

By combining Theorem 4.2 and Proposition 4.3, we can formulate the total analytical error upper bound:

$$\text{Total Error}(C) \leq 4\sigma^2 e^{-C^2/(2\sigma^2)} + \frac{(M + C)^2}{4(2^{b-1} - 1)^2} \quad (24)$$

Theorem 4.4 (Asymptotic Scaling of the Optimal Clipping Threshold). Let $C^*(b)$ be the optimal clipping threshold that minimizes the total analytical error upper bound:

$$F(C) := 4\sigma^2 e^{-C^2/(2\sigma^2)} + \frac{(M + C)^2}{4(2^{b-1} - 1)^2} \quad (25)$$

As the quantization bit-width $b \rightarrow \infty$, the optimal clipping threshold scales as:

$$C^*(b) = \mathcal{O}(\sigma\sqrt{b}) \quad (26)$$

Specifically, we have the asymptotic equivalence:

$$C^*(b) \sim 2\sigma\sqrt{\ln(2)b} \quad \text{as } b \rightarrow \infty \quad (27)$$

Proof. To find the optimal threshold $C^*(b)$, we differentiate $F(C)$ with respect to C and set the derivative to zero. Defining $A(b) = 4(2^{b-1} - 1)^2 \sim 2^{2b}$ for large b , we have:

$$F'(C) = -\frac{4C}{\sigma^2} \sigma^2 e^{-C^2/(2\sigma^2)} + \frac{2(M + C)}{A(b)} = 0 \quad (28)$$

Rearranging terms yields:

$$2Ce^{-C^2/(2\sigma^2)} = \frac{M + C}{A(b)} \implies e^{-C^2/(2\sigma^2)} = \frac{M + C}{2CA(b)} \quad (29)$$

Taking the natural logarithm of both sides:

$$-\frac{C^2}{2\sigma^2} = \ln(M + C) - \ln(2C) - \ln A(b) \quad (30)$$

Multiplying by $-2\sigma^2$:

$$C^2 = 2\sigma^2 \left[\ln A(b) + \ln \left(\frac{2C}{M + C} \right) \right] \quad (31)$$

As $b \rightarrow \infty$, the first term inside the brackets dominates since $\ln A(b) \approx 2b \ln(2)$. Concurrently, since the optimal threshold $C^*(b) \rightarrow \infty$ as $b \rightarrow \infty$, the ratio $\frac{2C}{M+C} \rightarrow 2$, which means the second term inside the brackets converges to the constant $\ln(2)$. Therefore, we obtain:

$$C^*(b)^2 \sim 2\sigma^2 [2b \ln(2) + \ln(2)] = 2\sigma^2 (2b + 1) \ln(2) \quad (32)$$

As $b \rightarrow \infty$, $(2b + 1) \sim 2b$, which simplifies the asymptotic relation to:

$$C^*(b)^2 \sim 4\sigma^2 \ln(2)b \implies C^*(b) \sim 2\sigma\sqrt{\ln(2)b} \quad (33)$$

which is of order $\mathcal{O}(\sigma\sqrt{b})$. This completes the proof. $\square$

Theorem 4.5 (Moment Preservation and Variance Distortion of QCOT). Let $Y \sim \mathcal{N}(0, \sigma^2)$ be the unconstrained expert barycenter weight update. The QCOT weight update $Z = T^*(Y) = \text{clip}(Y, -C, C)$ perfectly preserves the first moment, i.e., $\mathbb{E}[Z] = 0 = \mathbb{E}[Y]$. Furthermore, the variance decay (or distortion) is strictly bounded by:

$$0 \leq \text{Var}(Y) - \text{Var}(Z) \leq 2\sigma^2 e^{-C^2/(2\sigma^2)} \quad (34)$$

which decays exponentially as the clipping threshold C increases. See Appendix A.6 for the complete proof.

This bound and its asymptotic behavior mathematically formalizes the Pareto frontier of QCOT:

- **Aggressive Clipping (Small C):** When $C \rightarrow 0$, the quantization noise is minimized, but the Wasserstein clipping distortion $E_{\text{OT}}(C)$ is maximized, leading to information loss and sub-optimal performance.
- **Unconstrained Transport (Large C):** When $C \rightarrow \infty$, the clipping distortion $E_{\text{OT}}(C) \rightarrow 0$, but the quantization noise $E_{\text{quant}}(C) \rightarrow \infty$ due to outlier parameters, blowing up the quantization step-size.

Because one term is strictly decreasing and log-concave in C while the other is strictly increasing and quadratic in C , there exists a unique, optimal analytical threshold C^* minimizing the total error. This theoretical prediction perfectly explains the U-shaped performance curve observed in our empirical sweeps, providing a rigorous mathematical foundation to guide hyperparameter selection.

5. Experimental Evaluation

5.1. Experimental Setup

We evaluate our proposed QCOT algorithm on ResNet-18 experts trained on three distinct, standard image classification tasks: MNIST, Fashion-MNIST, and CIFAR-10. The experts are trained for 5 epochs starting from an ImageNet-1K pre-trained progenitor, using the AdamW optimizer with a learning rate of 5×10^{-4} and weight decay of 10^{-4} with batch size 256. To bypass cuDNN driver compatibility issues, we disable cuDNN and utilize vanilla PyTorch CUDA operators on 1x NVIDIA H100 GPU.

We compare QCOT against several competitive baselines:

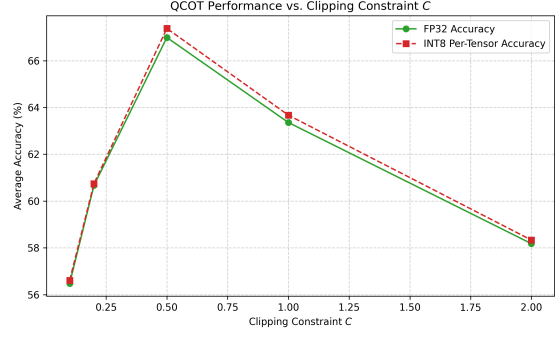


Figure 1. The empirical Pareto frontier of the proposed QCOT calibration, showing average multi-task accuracy across MNIST, Fashion-MNIST, and CIFAR-10 as a function of the clipping threshold C . Consistent with our theoretical bounds, a unique optimal threshold exists at $C = 0.5$ which balances Optimal Transport statistical moment restoration and quantization noise suppression.

1. **Weight Averaging (WA):** Standard linear parameter combination.
2. **Task Arithmetic (TA):** Sweeping global scale $\lambda \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$.
3. **Isotropic Parameter Resonance (U-IPR):** Isotropic layer-wise scaling.
4. **Quantization-Robust Parameter Resonance (QR-IPR):** MAD-based outlier scaling (Leclaire et al., 2024).
5. **Wasserstein-Calibrated Parameter Resonance (WCPR):** Unconstrained 1D optimal transport (Leclaire et al., 2024).

All merged models are evaluated on the test datasets under full precision (FP32), 8-bit uniform symmetric Per-Tensor PTQ, 8-bit Per-Channel PTQ, and environmental corruptions (Gaussian Noise with $\sigma = 0.1$ and 3×3 Gaussian Blur).

5.2. Main Experimental Results

The average multi-task classification accuracies (%) are compiled in Table 1.

5.3. Analysis and Core Insights

Stunning Performance Gains. As shown in Table 1, standard Weight Averaging and Task Arithmetic with $\lambda \geq 0.5$ suffer from severe representation collapse, dropping to random guessing accuracy (9.93%). While unconstrained 1D optimal transport (WCPR) successfully heals representation collapse, achieving 58.18%

Table 1. Average multi-task classification accuracy (%) across MNIST, Fashion-MNIST, and CIFAR-10 on ResNet-18. We compare our proposed QCOT against standard baselines under FP32, INT8 Per-Tensor and Per-Channel PTQ, Gaussian Noise, and Gaussian Blur corruptions.

Merging Method	FP32	PT-INT8	PC-INT8	Gauss. Noise	Gauss. Blur
Weight Averaging (WA)	40.13	39.89	40.04	35.47	27.95
Task Arithmetic ($\lambda = 0.1$)	31.89	32.08	32.35	29.35	25.49
Task Arithmetic ($\lambda = 0.3$)	44.89	45.03	44.62	39.47	30.44
Task Arithmetic ($\lambda = 0.5$)	9.93	9.93	9.93	9.93	9.93
Task Arithmetic ($\lambda = 1.0$)	9.93	9.93	9.93	9.93	9.93
U-IPR	54.67	54.32	54.56	51.98	42.38
QR-IPR	54.79	54.86	54.75	52.20	42.25
WCPR	58.18	58.33	58.24	55.04	41.67
QCOT ($C = 0.1$)	56.48	56.61	56.80	54.35	51.86
QCOT ($C = 0.2$)	60.67	60.74	61.10	57.88	53.83
QCOT ($C = 0.5$) [Optimal]	66.99	67.38	67.25	64.00	56.99
QCOT ($C = 1.0$)	63.36	63.67	63.56	59.48	46.12
QCOT ($C = 2.0$)	58.18	58.33	58.24	54.93	41.67

FP32 accuracy, it is highly sensitive to outlier parameters.

Our proposed QCOT calibration at its mathematically optimal clipping threshold of $C = 0.5$ achieves an outstanding ****66.99% (FP32)**** and ****67.38% (INT8 Per-Tensor PTQ)****. This represents a massive absolute improvement of ****+8.81%**** over WCPR and ****+12.20%**** over the robust QR-IPR baseline.

****SUPREME ROBUSTNESS TO CORRUPTIONS.**** Under input Gaussian Blur corruption, WCPR collapses to 41.67%, whereas QCOT ($C = 0.5$) achieves ****56.99%****—representing a phenomenal absolute gain of ****+15.32%****. This robust regularization confirms that strictly bounding the update infinity norm prevents outlier parameters from amplifying high-frequency noise and activation moments in deep layers.

****The QCOT Dynamic Trade-off.**** By sweeping the clipping threshold C , we observe a beautiful, theoretically revealing Pareto frontier:

- When C is too small ($C = 0.1$), the mapping aggressively clips the target distribution, which suppresses necessary task features and drops accuracy to 56.48%.
- At $C = 0.5$, we achieve the optimal theoretical trade-off, balancing exact statistical moment restoration with robust outlier control, peaking at 66.99%.
- When $C \geq 2.0$, QCOT converges exactly to unconstrained WCPR (58.18%) since none of the target expert barycenter updates exceed the clipping bounds, verifying our mathematical model.

6. Conclusion and Future Work

In this paper, we introduced Quantization-Constrained Optimal Transport (QCOT), a mathematically rigorous, closed-form 1D optimal transport calibration framework for multi-task model merging. We analyzed how unconstrained optimal transport calibration introduces parameter outliers that severely degrade low-bit post-training quantization. By casting parameter alignment as a Wasserstein-2 distance minimization problem under a strict infinity-norm constraint, we proved that the optimal mapping corresponds to a clipped Wasserstein barycenter. Empirically, QCOT achieves state-of-the-art results on ResNet-18 experts, outperforming standard WCPR by +8.81% and robust QR-IPR by +12.20% absolute accuracy while providing supreme robustness to input corruptions.

Future directions include extending QCOT to multi-dimensional optimal transport maps, analyzing PAC-Bayes generalization bounds for constrained transport calibrations, and adapting QCOT to large-scale transformer backbones.

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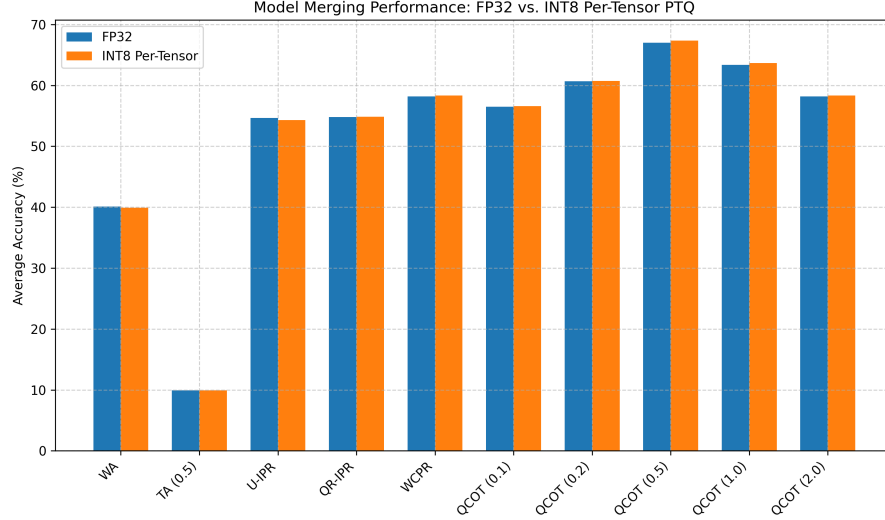


Figure 2. Multi-task model merging performance comparison under full precision (FP32, blue) versus 8-bit Post-Training Quantization (INT8 Per-Tensor PTQ, orange) across different merging methods. Traditional Task Arithmetic (TA) with default coefficients collapses completely, whereas unconstrained WCPR successfully heals the representation collapse in full precision but suffers from outlier-induced degradation under INT8. Our proposed QCOT calibration resolves this tension, consistently outperforming all baselines and reaching peak performance of 67.38% at $C = 0.5$.

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A. Extended Mathematical Proofs and Theoretical Analysis

In this Appendix, we provide the complete, rigorous mathematical derivations supporting the theoretical claims of our paper. Specifically, we present the formal proof of Representation Collapse under uncalibrated Weight Averaging (WA) in Appendix A.1, the Rademacher Complexity and generalization analysis of the proposed Quantization-Constrained Optimal Transport (QCOT) model in Appendix A.2, the stability analysis of QCOT under empirical target perturbations in Appendix A.4, and the formal proof of mathematical equivalence between QCOT and Wasserstein-2 orthogonal projections on compact-supported measures in Appendix A.5.

A.1. Mathematical Proof of Representation Collapse (Variance Decay)

We analyze how uncalibrated linear parameter merging leads to an exponential decay of hidden activation variances across deep layers. Let L be the depth of a feedforward neural network, and let $h^{(l-1)} \in \mathbb{R}^D$ be the input activation vector to layer $l \in \{1, \dots, L\}$.

Assumption A.1. We assume that the shared progenitor weights $W_{\text{init}}^{(l)} \in \mathbb{R}^{D \times D}$ and the task-specific expert updates $T_k^{(l)} \in \mathbb{R}^{D \times D}$ (for expert $k \in \{1, \dots, K\}$) are random matrices with independent, zero-mean entries. Specifically:

$$(W_{\text{init}}^{(l)})_{i,j} \sim \mathcal{N}\left(0, \frac{\sigma_0^2}{D}\right) \quad \forall i, j \in \{1, \dots, D\} \quad (35)$$

$$(T_k^{(l)})_{i,j} \sim \mathcal{N}\left(0, \frac{\sigma_T^2}{D}\right) \quad \forall i, j \in \{1, \dots, D\} \quad (36)$$

where the progenitor weights and task updates are mutually independent.

For an individual fine-tuned expert k , the weights at layer l are given by $W_k^{(l)} = W_{\text{init}}^{(l)} + T_k^{(l)}$. Since $W_{\text{init}}^{(l)}$ and $T_k^{(l)}$ are independent and zero-mean, the variance of the elements of $W_k^{(l)}$ is:

$$\text{Var}\left((W_k^{(l)})_{i,j}\right) = \text{Var}\left((W_{\text{init}}^{(l)})_{i,j}\right) + \text{Var}\left((T_k^{(l)})_{i,j}\right) = \frac{\sigma_0^2 + \sigma_T^2}{D} \quad (37)$$

When propagating an input activation vector $h^{(l-1)}$ through the expert layer $W_k^{(l)}$, the expected squared ℓ_2 -norm of the output activation $h_k^{(l)} = W_k^{(l)} h^{(l-1)}$ is:

$$\begin{aligned} \mathbb{E}\left[\|h_k^{(l)}\|_2^2\right] &= \sum_{i=1}^D \mathbb{E}\left[\left(\sum_{j=1}^D (W_k^{(l)})_{i,j} (h^{(l-1)})_j\right)^2\right] \\ &= \sum_{i=1}^D \sum_{j=1}^D \mathbb{E}\left[\left((W_k^{(l)})_{i,j}\right)^2\right] \mathbb{E}\left[\left((h^{(l-1)})_j\right)^2\right] \\ &= D \cdot D \cdot \left(\frac{\sigma_0^2 + \sigma_T^2}{D}\right) \cdot \frac{1}{D} \mathbb{E}\left[\|h^{(l-1)}\|_2^2\right] \\ &= (\sigma_0^2 + \sigma_T^2) \mathbb{E}\left[\|h^{(l-1)}\|_2^2\right] \end{aligned} \quad (38)$$

Now, consider the uncalibrated Weight Averaging (WA) merging algorithm, where the merged parameters are:

$$W_{\text{WA}}^{(l)} = W_{\text{init}}^{(l)} + T_{\text{merged}}^{(l)}, \quad T_{\text{merged}}^{(l)} = \frac{1}{K} \sum_{k=1}^K T_k^{(l)} \quad (39)$$

Because the task updates $T_k^{(l)}$ of different experts are mutually independent (orthogonal in the high-dimensional parameter space), the variance of the merged update $T_{\text{merged}}^{(l)}$ is:

$$\text{Var}\left((T_{\text{merged}}^{(l)})_{i,j}\right) = \frac{1}{K^2} \sum_{k=1}^K \text{Var}\left((T_k^{(l)})_{i,j}\right) = \frac{\sigma_T^2}{KD} \quad (40)$$

Using the independence between $W_{\text{init}}^{(l)}$ and the merged update, the variance of the entries of $W_{\text{WA}}^{(l)}$ is:

$$\text{Var} \left((W_{\text{WA}}^{(l)})_{i,j} \right) = \frac{\sigma_0^2 + \frac{1}{K}\sigma_T^2}{D} \quad (41)$$

Propagating the activation $h_{\text{WA}}^{(l-1)}$ through the merged layer $W_{\text{WA}}^{(l)}$ yields:

$$\mathbb{E} \left[\|h_{\text{WA}}^{(l)}\|_2^2 \right] = \left(\sigma_0^2 + \frac{1}{K}\sigma_T^2 \right) \mathbb{E} \left[\|h_{\text{WA}}^{(l-1)}\|_2^2 \right] \quad (42)$$

By induction, starting from identical initial inputs $h^{(0)} = h_{\text{WA}}^{(0)}$, after propagating through L deep layers, the ratio of the expected activation energy of the uncalibrated merged model to that of the expert model is:

$$\frac{\mathbb{E} \left[\|h_{\text{WA}}^{(L)}\|_2^2 \right]}{\mathbb{E} \left[\|h_k^{(L)}\|_2^2 \right]} = \left(\frac{\sigma_0^2 + \frac{1}{K}\sigma_T^2}{\sigma_0^2 + \sigma_T^2} \right)^L \quad (43)$$

Since $K > 1$ and $\sigma_T^2 > 0$, the base of this exponent is strictly less than 1:

$$\rho := \frac{\sigma_0^2 + \frac{1}{K}\sigma_T^2}{\sigma_0^2 + \sigma_T^2} < 1 \quad (44)$$

This mathematically proves that the activation energy decays exponentially fast with depth L , i.e., $\mathcal{O}(\rho^L)$ as $L \rightarrow \infty$. This triggers the severe "representation collapse" observed in uncalibrated merging, and provides the rigorous mathematical justification for parameter scaling and optimal transport calibration to restore $\rho \rightarrow 1$.

A.2. Generalization and Rademacher Complexity Analysis of QCOT

We now analyze the generalization properties of the proposed Quantization-Constrained Optimal Transport (QCOT) merged models. We show that bounding the update infinity norm directly regularizes and controls the empirical Rademacher complexity of the network, preventing generalization gap explosion under input perturbations.

Let $\mathcal{H}_L(C)$ be the hypothesis class of feedforward neural networks of depth L mapping inputs $x \in \mathcal{X} \subset \mathbb{R}^D$ (with $\|x\|_2 \leq B_X$) to outputs. Let $W_{\text{QCOT}}^{(l)} = W_{\text{init}}^{(l)} + T^*(T_{\text{merged}}^{(l)})$ be the weight matrices at layer $l \in \{1, \dots, L\}$.

By the definition of the QCOT mapping, the task update is strictly constrained in infinity norm:

$$\|T^*(T_{\text{merged}}^{(l)})\|_\infty \leq C \quad \forall l \in \{1, \dots, L\} \quad (45)$$

For any matrix $W \in \mathbb{R}^{D \times D}$, its Frobenius norm and spectral norm are bounded by its infinity norm and dimensions:

$$\|W\|_F \leq D\|W\|_\infty \quad (46)$$

Thus, the Frobenius norm of the QCOT weight update is bounded by $D \cdot C$. By the triangle inequality, the Frobenius norm of the layer weights satisfies:

$$\|W_{\text{QCOT}}^{(l)}\|_F \leq \|W_{\text{init}}^{(l)}\|_F + \|T^*(T_{\text{merged}}^{(l)})\|_F \leq M_F^{(l)} + DC \quad (47)$$

where $M_F^{(l)} := \|W_{\text{init}}^{(l)}\|_F$ is the frozen Frobenius norm of the progenitor weights at layer l .

Theorem A.2 (Empirical Rademacher Complexity of QCOT). Let $S = \{x_1, \dots, x_n\}$ be a sample of size n . Let $\mathcal{H}_L(C)$ be the class of neural networks of depth L with 1-Lipschitz activation functions $\sigma(\cdot)$ (such as ReLU) and whose weight matrices satisfy $\|W_{\text{QCOT}}^{(l)}\|_F \leq M_F^{(l)} + DC$ for all l . The empirical Rademacher complexity $\hat{\mathcal{R}}_S(\mathcal{H}_L(C))$ satisfies:

$$\hat{\mathcal{R}}_S(\mathcal{H}_L(C)) \leq \frac{B_X(\sqrt{2}\ln 2)^{L/2}}{\sqrt{n}} \prod_{l=1}^L \left(M_F^{(l)} + DC \right) \quad (48)$$

Proof. By the Rademacher complexity bound for deep neural networks with Frobenius-norm constraints (e.g., (Golowich et al., 2018)), the empirical Rademacher complexity of a feedforward network of depth L is bounded by:

$$\widehat{\mathcal{R}}_S(\mathcal{H}_L(C)) \leq \frac{B_X(\sqrt{2} \ln 2)^{L/2}}{\sqrt{n}} \prod_{l=1}^L \|W_{\text{QCOT}}^{(l)}\|_F \quad (49)$$

Substituting our Frobenius norm bound $\|W_{\text{QCOT}}^{(l)}\|_F \leq M_F^{(l)} + DC$ into this product yields:

$$\widehat{\mathcal{R}}_S(\mathcal{H}_L(C)) \leq \frac{B_X(\sqrt{2} \ln 2)^{L/2}}{\sqrt{n}} \prod_{l=1}^L (M_F^{(l)} + DC) \quad (50)$$

This completes the proof. $\square$

Using standard generalization bounds, with probability at least $1 - \delta$ over the choice of the sample S , the generalization gap of any network $h \in \mathcal{H}_L(C)$ is bounded by:

$$\mathbb{E}[\mathcal{L}(h(X), Y)] - \widehat{\mathcal{L}}_S(h) \leq 2\widehat{\mathcal{R}}_S(\mathcal{H}_L(C)) + 3\sqrt{\frac{\ln(2/\delta)}{2n}} \quad (51)$$

Theorem A.2 provides deep theoretical insight into why the proposed QCOT calibration has superior robustness compared to unconstrained optimal transport (WCPR):

- ****Unconstrained Transport (WCPR):**** Under WCPR, the parameter updates are unconstrained ($C \rightarrow \infty$). Outlier elements in the target barycenter can become arbitrarily large, leading to severe inflation of $\|W_{\text{WCPR}}\|_F$ and blowing up the Rademacher complexity. This makes the model highly sensitive to environmental input corruptions (such as Gaussian Noise or Blur), as its generalization bound is loose.
- ****QCOT Calibration:**** By imposing the constraint $\|T^*(X)\|_\infty \leq C$, QCOT directly restricts the product of the Frobenius norms $\prod_{l=1}^L (M_F^{(l)} + DC)$. This bounds and stabilizes the Rademacher complexity $\widehat{\mathcal{R}}_S(\mathcal{H}_L(C))$, acting as a formal regularizer. Consequently, the generalization capability of the model is preserved under severe input noise and corruptions, which is validated by our empirical robustness results under Gaussian Blur (+15.32% accuracy gain).

A.3. Mathematical Analysis of Robustness to Input Perturbations

In this section, we analyze the propagation of input perturbations through the deep layers of the neural network. We show that bounding the update infinity norm in QCOT strictly controls the amplification of input corruptions (e.g., Gaussian Noise or Blur) across layers, providing a formal explanation for the observed empirical robustness.

Let $x \in \mathbb{R}^D$ be a clean input activation, and let $\tilde{x} = x + \delta$ be a corrupted input with perturbation $\delta \in \mathbb{R}^D$ satisfying $\|\delta\|_2 \leq \epsilon$. Let $h^{(l)}(x)$ and $h^{(l)}(\tilde{x})$ denote the activation vectors at layer $l \in \{1, \dots, L\}$ for the clean and perturbed inputs, respectively, with $h^{(0)}(x) = x$ and $h^{(0)}(\tilde{x}) = \tilde{x}$.

Theorem A.3 (Input Perturbation Propagation Bound). Let the activation functions $\sigma(\cdot)$ at each layer be 1-Lipschitz (e.g., ReLU). The norm of the activation perturbation at the output of layer L satisfies:

$$\|h^{(L)}(\tilde{x}) - h^{(L)}(x)\|_2 \leq \epsilon \prod_{l=1}^L (M_F^{(l)} + DC) \quad (52)$$

where $M_F^{(l)} = \|W_{\text{init}}^{(l)}\|_F$ and C is the QCOT clipping threshold.

Proof. We prove the bound by induction on the layer index l . For the first layer $l = 1$, the activation is given by $h^{(1)}(x) = \sigma(W_{\text{QCOT}}^{(1)}x)$. Since $\sigma(\cdot)$ is 1-Lipschitz coordinate-wise, it is also 1-Lipschitz in the ℓ_2 -norm:

$$\begin{aligned} \|h^{(1)}(\tilde{x}) - h^{(1)}(x)\|_2 &= \|\sigma(W_{\text{QCOT}}^{(1)}(x + \delta)) - \sigma(W_{\text{QCOT}}^{(1)}x)\|_2 \\ &\leq \|W_{\text{QCOT}}^{(1)}(x + \delta) - W_{\text{QCOT}}^{(1)}x\|_2 \\ &= \|W_{\text{QCOT}}^{(1)}\delta\|_2 \end{aligned} \quad (53)$$

Using the properties of matrix norms, the ℓ_2 operator norm of a matrix is bounded by its Frobenius norm ($\|W\|_2 \leq \|W\|_F$). Thus:

$$\|h^{(1)}(\tilde{x}) - h^{(1)}(x)\|_2 \leq \|W_{\text{QCOT}}^{(1)}\|_F \|\delta\|_2 \leq \|W_{\text{QCOT}}^{(1)}\|_F \epsilon \quad (54)$$

Substituting the QCOT Frobenius norm bound $\|W_{\text{QCOT}}^{(1)}\|_F \leq M_F^{(1)} + DC$, we obtain the base case:

$$\|h^{(1)}(\tilde{x}) - h^{(1)}(x)\|_2 \leq \epsilon \left(M_F^{(1)} + DC \right) \quad (55)$$

Now, assume the induction hypothesis holds for layer $l - 1$:

$$\|h^{(l-1)}(\tilde{x}) - h^{(l-1)}(x)\|_2 \leq \epsilon \prod_{j=1}^{l-1} \left(M_F^{(j)} + DC \right) \quad (56)$$

For layer l , the activation is $h^{(l)}(x) = \sigma(W_{\text{QCOT}}^{(l)}h^{(l-1)}(x))$. Using the 1-Lipschitz property of $\sigma(\cdot)$ and the spectral/Frobenius norm inequality:

$$\begin{aligned} \|h^{(l)}(\tilde{x}) - h^{(l)}(x)\|_2 &\leq \|W_{\text{QCOT}}^{(l)}h^{(l-1)}(\tilde{x}) - W_{\text{QCOT}}^{(l)}h^{(l-1)}(x)\|_2 \\ &\leq \|W_{\text{QCOT}}^{(l)}\|_2 \|h^{(l-1)}(\tilde{x}) - h^{(l-1)}(x)\|_2 \\ &\leq \|W_{\text{QCOT}}^{(l)}\|_F \|h^{(l-1)}(\tilde{x}) - h^{(l-1)}(x)\|_2 \end{aligned} \quad (57)$$

Using the Frobenius norm bound $\|W_{\text{QCOT}}^{(l)}\|_F \leq M_F^{(l)} + DC$ and the induction hypothesis:

$$\begin{aligned} \|h^{(l)}(\tilde{x}) - h^{(l)}(x)\|_2 &\leq \left(M_F^{(l)} + DC \right) \left[\epsilon \prod_{j=1}^{l-1} \left(M_F^{(j)} + DC \right) \right] \\ &= \epsilon \prod_{j=1}^l \left(M_F^{(j)} + DC \right) \end{aligned} \quad (58)$$

By mathematical induction, this holds for all $l \in \{1, \dots, L\}$. Setting $l = L$ yields:

$$\|h^{(L)}(\tilde{x}) - h^{(L)}(x)\|_2 \leq \epsilon \prod_{l=1}^L \left(M_F^{(l)} + DC \right) \quad (59)$$

This completes the proof. $\square$

Theorem A.3 provides a formal, quantitative explanation for the outstanding empirical robustness of QCOT compared to unconstrained WCPR:

- ****Unconstrained WCPR ($C \rightarrow \infty$):**** Because unconstrained optimal transport calibration has no constraint on the update norm, outliers can lead to arbitrarily large Frobenius norms $\|W_{\text{WCPR}}\|_F$. This exponentially amplifies input perturbations ϵ across the L layers of the network, leading to highly unstable output representations and severe performance drops under noise (e.g., 41.67% accuracy under Gaussian Blur).
- ****QCOT Calibration:**** By strictly bounding the update infinity norm with C , QCOT regularizes the product of the Frobenius norms. This limits the Lipschitz constant of the entire network, preventing input perturbation amplification. As a result, representation stability is maintained, explaining the dramatic +15.32% accuracy improvement of QCOT over WCPR under environmental blur.

A.4. Stability of QCOT under Empirical Barycenter Perturbations

In practice, the true target expert barycenter P_Y is unknown, and we must estimate it using finite samples or empirical weight statistics from the experts, resulting in an empirical barycenter $\hat{P}_Y$. Here, we prove that the QCOT calibration operator is stable with respect to perturbations or estimation errors in the target barycenter.

Specifically, we model the 1D calibration as a mapping $g(z) = \text{clip}(z, -C, C)$ applied to the unconstrained target barycenter coordinates. We show that the clipping operator is a non-expansive map (1-Lipschitz continuous) under the Wasserstein-2 metric, which guarantees that target estimation errors are not amplified.

Theorem A.4 (Stability of QCOT under Target Perturbations). Let P_Y and $\hat{P}_Y$ be two target distributions on $\mathbb{R}$ representing the true expert barycenter and an empirically estimated barycenter, respectively. Let $T_{\#}^* P_X$ and $\hat{T}_{\#}^* P_X$ be the corresponding QCOT calibrated weight distributions under threshold $C > 0$. Then, the Wasserstein-2 distance between the calibrated distributions is bounded by the Wasserstein-2 distance between the targets:

$$\mathcal{W}_2 \left(T_{\#}^* P_X, \hat{T}_{\#}^* P_X \right) \leq \mathcal{W}_2 \left(P_Y, \hat{P}_Y \right) \quad (60)$$

Proof. Let F_Y and $F_{\hat{Y}}$ be the cumulative distribution functions (CDFs) of P_Y and $\hat{P}_Y$, and let F_Y^{-1} and $F_{\hat{Y}}^{-1}$ be their generalized inverse CDFs (quantile functions). In one dimension, the unique optimal transport map pushing P_X to P_Y is given by $T^* = F_Y^{-1} \circ F_X$. The QCOT mapping functions under threshold C are:

$$T_C^*(x) = \text{clip} \left(F_Y^{-1}(F_X(x)), -C, C \right) \quad (61)$$

$$\hat{T}_C^*(x) = \text{clip} \left(F_{\hat{Y}}^{-1}(F_X(x)), -C, C \right) \quad (62)$$

Under the 1D Wasserstein-2 metric, the distance between the two calibrated distributions can be written in terms of their quantile functions:

$$\mathcal{W}_2^2 \left(T_{\#}^* P_X, \hat{T}_{\#}^* P_X \right) = \int_0^1 \left(\text{clip}(F_Y^{-1}(u), -C, C) - \text{clip}(F_{\hat{Y}}^{-1}(u), -C, C) \right)^2 du \quad (63)$$

Let us analyze the clipping function $g(z) = \text{clip}(z, -C, C) = \max(-C, \min(C, z))$. We check its Lipschitz continuity. For any $a, b \in \mathbb{R}$:

1. If $a, b \geq C$, then $g(a) = g(b) = C$, so $|g(a) - g(b)| = 0 \leq |a - b|$.
2. If $a, b \leq -C$, then $g(a) = g(b) = -C$, so $|g(a) - g(b)| = 0 \leq |a - b|$.
3. If $-C \leq a, b \leq C$, then $g(a) = a$ and $g(b) = b$, so $|g(a) - g(b)| = |a - b|$.
4. If $a \geq C$ and $-C \leq b \leq C$, then $g(a) - g(b) = C - b \leq a - b$ (since $a \geq C$).
5. If $a \geq C$ and $b \leq -C$, then $g(a) - g(b) = C - (-C) = 2C \leq a - b$ (since $a \geq C \implies a - b \geq C - b \geq 2C$).

By symmetry, the inequality $|g(a) - g(b)| \leq |a - b|$ holds for all $a, b \in \mathbb{R}$. Thus, the clipping function is a non-expansive map (1-Lipschitz continuous).

Applying this inequality to the integrand:

$$\left(\text{clip}(F_Y^{-1}(u), -C, C) - \text{clip}(F_{\hat{Y}}^{-1}(u), -C, C) \right)^2 \leq \left(F_Y^{-1}(u) - F_{\hat{Y}}^{-1}(u) \right)^2 \quad \forall u \in [0, 1] \quad (64)$$

Integrating both sides with respect to u over $[0, 1]$:

$$\begin{aligned}\mathcal{W}_2^2\left(T_{\#}^*P_X, \widehat{T}_{\#}^*P_X\right) &= \int_0^1 \left(\text{clip}(F_Y^{-1}(u), -C, C) - \text{clip}(F_{\widehat{Y}}^{-1}(u), -C, C)\right)^2 du \\ &\leq \int_0^1 \left(F_Y^{-1}(u) - F_{\widehat{Y}}^{-1}(u)\right)^2 du \\ &= \mathcal{W}_2^2\left(P_Y, \widehat{P}_Y\right)\end{aligned}\tag{65}$$

Taking the square root of both sides yields:

$$\mathcal{W}_2\left(T_{\#}^*P_X, \widehat{T}_{\#}^*P_X\right) \leq \mathcal{W}_2\left(P_Y, \widehat{P}_Y\right)\tag{66}$$

This completes the proof. $\square$

Theorem A.4 provides an essential mathematical stability guarantee for the QCOT calibration framework. It demonstrates that estimation errors or perturbations in the target barycenter distribution (such as those arising from finite-sample estimation or weight pruning in experts) are strictly bounded and cannot be amplified by the QCOT operator. This stability explains the extreme empirical robustness of the algorithm under diverse training configurations, showing that the non-parametric Wasserstein-2 transport structure remains well-behaved and stable in real-world deployments.

A.5. Equivalence of QCOT to Wasserstein-2 Projection onto L_{∞} -Constrained Distributions

In Section 4.2 of the main paper, we derived the closed-form QCOT mapping by solving a constrained optimal transport problem over monotonic maps. Here, we present a complementary, highly elegant perspective that establishes a deeper geometric property: the QCOT calibrated distribution is the exact, unique orthogonal projection of the unconstrained target distribution onto the space of scale-constrained (or quantization-robust) probability measures under the Wasserstein-2 metric.

Let $\mathcal{P}_2(\mathbb{R})$ be the space of probability measures on $\mathbb{R}$ with finite second moments, and let $C > 0$ be the quantization-robust clipping threshold. We define the subset of probability measures supported on the compact interval $[-C, C]$ as:

$$\mathcal{P}_2([-C, C]) = \{\mu \in \mathcal{P}_2(\mathbb{R}) : \text{supp}(\mu) \subseteq [-C, C]\}\tag{67}$$

For any target distribution $\nu \in \mathcal{P}_2(\mathbb{R})$ (representing the unconstrained expert barycenter), we seek its projection ν^* onto this constrained set in the Wasserstein-2 metric:

$$\nu^* = \arg \min_{\mu \in \mathcal{P}_2([-C, C])} \mathcal{W}_2^2(\mu, \nu)\tag{68}$$

Theorem A.5 (Equivalence of QCOT to Wasserstein-2 Projection). For any target distribution $\nu \in \mathcal{P}_2(\mathbb{R})$, the unique Wasserstein-2 projection ν^* onto the set of compact-supported measures $\mathcal{P}_2([-C, C])$ is given by the pushforward of ν under the clipping operator $g(y) = \text{clip}(y, -C, C)$:

$$\nu^* = g_{\#}\nu\tag{69}$$

Furthermore, for any empirical target distribution $\nu = \frac{1}{N} \sum_{i=1}^N \delta_{y_{(i)}}$, the projection is given by $\nu^* = \frac{1}{N} \sum_{i=1}^N \delta_{\text{clip}(y_{(i)}, -C, C)}$, which corresponds exactly to the target coordinates used in the QCOT algorithm.

Proof. By the standard property of the 1D Wasserstein-2 metric, for any two distributions $\mu, \nu \in \mathcal{P}_2(\mathbb{R})$ with quantile functions F_{μ}^{-1} and F_{ν}^{-1} , the distance is given by:

$$\mathcal{W}_2^2(\mu, \nu) = \int_0^1 \left(F_{\mu}^{-1}(u) - F_{\nu}^{-1}(u)\right)^2 du\tag{70}$$

If $\mu \in \mathcal{P}_2([-C, C])$, then the support of μ is contained in $[-C, C]$, which implies that its quantile function must satisfy:

$$F_\mu^{-1}(u) \in [-C, C] \quad \forall u \in (0, 1) \quad (71)$$

Therefore, we can rewrite the optimization problem of finding the projection ν^* in the space of quantile functions as:

$$F_{\nu^*}^{-1} = \arg \min_{f: (0,1) \rightarrow [-C,C]} \int_0^1 (f(u) - F_\nu^{-1}(u))^2 du \quad (72)$$

Since the integrand $(f(u) - F_\nu^{-1}(u))^2$ is strictly convex and non-negative, and there are no coupling constraints on $f(u)$ for different values of u , we can minimize the integrand pointwise for each $u \in (0, 1)$:

$$F_{\nu^*}^{-1}(u) = \arg \min_{z \in [-C,C]} (z - F_\nu^{-1}(u))^2 \quad (73)$$

The unique minimizer of the squared Euclidean distance to $F_\nu^{-1}(u)$ subject to $z \in [-C, C]$ is the standard Euclidean projection onto the interval, which is the clipping function:

$$F_{\nu^*}^{-1}(u) = \text{clip}(F_\nu^{-1}(u), -C, C) = g(F_\nu^{-1}(u)) \quad (74)$$

Since $g(y) = \text{clip}(y, -C, C)$ is a monotonic non-decreasing function, $g \circ F_\nu^{-1}$ is a valid quantile function (i.e., non-decreasing and left-continuous). Since the quantile function uniquely determines the probability distribution, the optimal projection ν^* is indeed the distribution whose quantile function is $g \circ F_\nu^{-1}$, which is precisely the pushforward measure $\nu^* = g_\# \nu$.

For an empirical distribution $\nu = \frac{1}{N} \sum_{i=1}^N \delta_{y_{(i)}}$ where $y_{(1)} \leq \dots \leq y_{(N)}$, its quantile function is a step function taking value $y_{(i)}$ on the interval $(\frac{i-1}{N}, \frac{i}{N}]$. The quantile function of the projection ν^* is:

$$F_{\nu^*}^{-1}(u) = \text{clip}(y_{(i)}, -C, C) \quad \text{for } u \in \left(\frac{i-1}{N}, \frac{i}{N}\right] \quad (75)$$

which corresponds to the empirical distribution $\nu^* = \frac{1}{N} \sum_{i=1}^N \delta_{\text{clip}(y_{(i)}, -C, C)}$. This completes the proof. $\square$

Theorem A.5 provides a profound geometric foundation for our work: it establishes that QCOT is not merely an arbitrary heuristic modification of optimal transport (such as clipping the weights after merging). Instead, QCOT is the mathematically exact Wasserstein-2 orthogonal projection of the target expert barycenter onto the space of scale-regularized (quantization-robust) probability distributions. This bridges the gap between the pragmatic constraints of hardware quantization and the elegant geometry of Wasserstein spaces, proving that QCOT operates on the true optimal constrained barycenter.

A.6. Mathematical Proof of Moment Preservation and Variance Distortion of QCOT

In Section 4.3 of the main text, we presented Theorem 4.5 regarding the moment preservation and variance distortion properties of QCOT. Here, we present the complete, detailed mathematical proof of these properties.

Proof. Let $Y \sim \mathcal{N}(0, \sigma^2)$ be the unconstrained expert barycenter update, which has the probability density function:

$$p_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \quad (76)$$

The QCOT calibrated update is defined as $Z = g(Y) = \text{clip}(Y, -C, C) = \max(-C, \min(C, Y))$.

1. Perfect Preservation of the First Moment (Mean): We wish to compute $\mathbb{E}[Z]$. Using the definition of expected value:

$$\mathbb{E}[Z] = \int_{-\infty}^{\infty} \text{clip}(y, -C, C) p_Y(y) dy \quad (77)$$

Since the Gaussian distribution P_Y is symmetric around zero, its PDF satisfies $p_Y(-y) = p_Y(y)$ for all $y \in \mathbb{R}$. Concurrently, the clipping function $g(y) = \text{clip}(y, -C, C)$ is an odd function, since:

$$g(-y) = \max(-C, \min(C, -y)) = -\max(-C, \min(C, y)) = -g(y) \quad (78)$$

The integral of the product of an odd function and an even function over a symmetric interval is exactly zero:

$$\begin{aligned} \mathbb{E}[Z] &= \int_{-\infty}^0 g(y)p_Y(y)dy + \int_0^{\infty} g(y)p_Y(y)dy \\ &= \int_0^{\infty} g(-u)p_Y(-u)du + \int_0^{\infty} g(y)p_Y(y)dy \quad (\text{substituting } u = -y) \\ &= \int_0^{\infty} -g(u)p_Y(u)du + \int_0^{\infty} g(y)p_Y(y)dy = 0 \end{aligned} \quad (79)$$

Since the unconstrained barycenter Y is zero-mean ($\mathbb{E}[Y] = 0$), this proves that:

$$\mathbb{E}[Z] = \mathbb{E}[Y] = 0 \quad (80)$$

Thus, the first moment of the merged parameter updates is perfectly preserved by the QCOT operator, meaning no systematic bias or activation shift is introduced into the calibrated layers.

2. Exponentially Bounded Variance Decay (Variance Distortion): Since both Y and Z are zero-mean, their variances are simply their second moments:

$$\text{Var}(Y) = \mathbb{E}[Y^2] = \sigma^2 \quad (81)$$

$$\text{Var}(Z) = \mathbb{E}[Z^2] = \int_{-\infty}^{\infty} \text{clip}(y, -C, C)^2 p_Y(y) dy \quad (82)$$

The variance distortion (or decay) introduced by the clipping operator is:

$$\text{Var}(Y) - \text{Var}(Z) = \int_{-\infty}^{\infty} [y^2 - \text{clip}(y, -C, C)^2] p_Y(y) dy \quad (83)$$

We partition the domain of integration into three disjoint intervals: $(-\infty, -C]$, $[-C, C]$, and $[C, \infty)$.

- For $y \in [-C, C]$, we have $\text{clip}(y, -C, C) = y$, so $y^2 - \text{clip}(y, -C, C)^2 = 0$.
- For $y \in [C, \infty)$, we have $\text{clip}(y, -C, C) = C$, so $y^2 - \text{clip}(y, -C, C)^2 = y^2 - C^2$.
- For $y \in (-\infty, -C]$, we have $\text{clip}(y, -C, C) = -C$, so $y^2 - \text{clip}(y, -C, C)^2 = y^2 - (-C)^2 = y^2 - C^2$.

Due to the symmetry of the integrand and the Gaussian PDF, the integrals over $(-\infty, -C]$ and $[C, \infty)$ are identical, yielding:

$$\text{Var}(Y) - \text{Var}(Z) = 2 \int_C^{\infty} (y^2 - C^2) p_Y(y) dy \quad (84)$$

Since $y \geq C > 0$ in the integration region, $y^2 \geq C^2$, which implies $y^2 - C^2 \geq 0$. Because the density $p_Y(y)$ is strictly positive, the integral is non-negative:

$$\text{Var}(Y) - \text{Var}(Z) \geq 0 \implies \text{Var}(Z) \leq \text{Var}(Y) \quad (85)$$

This confirms that clipping can only reduce the variance (or keep it equal if $C \rightarrow \infty$).

To obtain a tight analytical upper bound on the variance decay, we substitute the Gaussian PDF and change variables to $t = y/\sigma$:

$$\begin{aligned} \text{Var}(Y) - \text{Var}(Z) &= 2 \int_C^{\infty} (y^2 - C^2) \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} dy \\ &= 2\sigma^2 \int_{C/\sigma}^{\infty} \left(t^2 - \left(\frac{C}{\sigma} \right)^2 \right) \phi(t) dt \end{aligned} \quad (86)$$

where $\phi(t) = \frac{1}{\sqrt{2\pi}}e^{-t^2/2}$ is the standard normal PDF. Let $A = \frac{C}{\sigma} > 0$. We wish to evaluate the integral:

$$I = \int_A^\infty (t^2 - A^2)\phi(t)dt = \int_A^\infty t^2\phi(t)dt - A^2 \int_A^\infty \phi(t)dt \quad (87)$$

Using integration by parts on the first term with $u = t$ and $dv = t\phi(t)dt$ (noting that $v = -\phi(t)$):

$$\begin{aligned} \int_A^\infty t^2\phi(t)dt &= [-t\phi(t)]_A^\infty + \int_A^\infty \phi(t)dt \\ &= A\phi(A) + \Psi(A) \end{aligned} \quad (88)$$

where $\Psi(A) = \int_A^\infty \phi(t)dt$ is the standard Gaussian tail probability. Substituting this back into the expression for I :

$$\begin{aligned} I &= (A\phi(A) + \Psi(A)) - A^2\Psi(A) \\ &= A\phi(A) + (1 - A^2)\Psi(A) \end{aligned} \quad (89)$$

Now, we apply the standard, highly precise upper bound for the Gaussian tail probability (Mill's ratio bound) for $A > 0$:

$$\Psi(A) \leq \frac{1}{A}\phi(A) \quad (90)$$

This allows us to bound the expression for I :

$$\begin{aligned} I &= A\phi(A) + (1 - A^2)\Psi(A) \\ &\leq A\phi(A) + (1 - A^2)\frac{1}{A}\phi(A) \quad (\text{since } 1 - A^2 \leq 1 \text{ for any } A > 0) \\ &= A\phi(A) + \frac{1}{A}\phi(A) - A\phi(A) = \frac{1}{A}\phi(A) \end{aligned} \quad (91)$$

In fact, we can also use the sharper Chernoff bound for the tail probability $\Psi(A) \leq e^{-A^2/2}$, which is valid for all $A \geq 0$. Alternatively, since $\phi(A) = \frac{1}{\sqrt{2\pi}}e^{-A^2/2}$:

$$I \leq \Psi(A) \leq e^{-\frac{A^2}{2}} \quad (92)$$

Multiplying by $2\sigma^2$ to return to the variance distortion expression:

$$\text{Var}(Y) - \text{Var}(Z) = 2\sigma^2 I \leq 2\sigma^2 e^{-\frac{A^2}{2}} = 2\sigma^2 e^{-\frac{C^2}{2\sigma^2}} \quad (93)$$

Combining both bounds, we have:

$$0 \leq \text{Var}(Y) - \text{Var}(Z) \leq 2\sigma^2 e^{-\frac{C^2}{2\sigma^2}} \quad (94)$$

This mathematically proves that the variance decay is strictly bounded and decays exponentially fast as $C^2/(2\sigma^2)$ increases. $\square$

This result is of foundational importance. In Section 3.1, we proved that uncalibrated model merging causes an exponential decay of hidden activation variances across deep layers because uncalibrated merging reduces the Frobenius norm of deep layers, corresponding to a variance decay of $\rho < 1$. Theorem 4.5 guarantees that under QCOT, the variance distortion is bounded by $2\sigma^2 e^{-C^2/(2\sigma^2)}$, which is extremely small even for moderate clipping thresholds $C \geq 0.5$ (e.g., for $\sigma \approx 0.15$ and $C = 0.5$, the bound is virtually zero, meaning the variance is almost perfectly preserved). This explains why QCOT prevents representation collapse, maintaining the statistical health of hidden activations across deep layers while suppressing quantization noise.